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Faculty of Information Technology

Department All

FINAL EXAM: **Academic year 2026-2027 (Semester I)**

Course Code and Name: INSY 8311 & Database Development with PL/SQL

Instructor: Eric Maniraguha
Time: 6:00 pm – 8:00 pm
Group: I, B, C and D
Exam Duration: 2.5 hours
Date: 13 Nov 2026, 14:00
Total max: Final Exam/40

Final Exams – Capstone Project based — Phase 0: Research, Data Source and Project Proposal

PHASE 0 — Research, Data Source and Project Proposal

Choose an original problem, research it, find real data, and present your plan

1. The project at a glance

The capstone is an **individual project and your Final Examination**. You will build a working Oracle database with PL/SQL for a real problem, using real data, in eight phases. Every phase is submitted through your GitHub repository.

Phase	Focus	Main deliverable
0	Research, data source, proposal	Research log, proposal, presentation, GitHub repo
I	Business process modeling	BPMN/UML swimlane diagram
II	Data sourcing and normalization	Raw data → 3NF tables, ERD, data dictionary
III	Database (PDB) creation	Pluggable database and OEM screenshots
IV	Tables and data loading	CREATE scripts, data loaded and validated
V	PL/SQL development	Procedures, functions, cursors, packages
VI	Advanced programming and auditing	Triggers, restriction rule, audit log
VII	Documentation, innovation, final presentation	Complete repository, innovation demo, 10 slides

How the project is graded (/40)

Category	What is assessed	Marks
Problem definition and modeling	Phases 0–I: problem, research, business process	6
Data and database design	Phases II–IV: data source, normalization (3NF), PDB, tables, data loading	10
PL/SQL programming and auditing	Phases V–VI: procedures, functions, cursors, packages, triggers, auditing	10
Documentation and presentation	Phase VII: GitHub repository, final presentation, defence	6
Innovation	An interactive layer on your database (see Innovation section)	8
Total		40

2. Choose your topic

- Choose a **real problem** in any area: health, agriculture, banking, logistics, mining, security, business, education, or another field you know.
- **Not allowed:** Student Management Systems, Hospital Management Systems and Employee Management Systems.
- **Your idea must not already be taken.** Open the registration sheet, read every topic already registered, and choose a different idea.
- Register your **full name, project title and a 2–3 sentence description** in the sheet below.

> **Register here:** https://docs.google.com/spreadsheets/d/1_1qbX-viAAwHAzxWR6G_f31yzB6ks7sV1-HgwAgbdVA/edit?usp=sharing

First come, first served: if your idea is already in the sheet, choose another. Registration is not approval; the instructor may ask you to change your topic.

Example project ideas (for inspiration only: check the sheet first)

Area	Project idea	Possible innovation
Agriculture	Cooperative crop input and harvest tracker	Power BI yield dashboard by season and district
Banking	SACCO micro-loan repayment and risk monitor	Risk-score function with a live dashboard
Logistics	Fleet delivery scheduling and fuel tracking	APEX app for dispatchers; triggers block double booking
Mining	Mine safety incidents and equipment inspections	Grafana alerts on overdue inspections
Public safety	Traffic violations and fine payments	Map of violation hotspots in Power BI
Utilities	Water meter readings and leak detection	Automated alert when usage jumps suddenly

3. Research the problem

Collect **at least 6 credible sources**: 2 research papers, 1 report or statistic, 1 existing system, 1 dataset (section 4) and 1 technical source. Search on Google Scholar, IEEE Xplore, ScienceDirect, NISR Rwanda, World Bank Data and Oracle Docs.

Record them in `docs/research/research_log.md`:

```
| # | Source (APA) | Type | Link | Key finding | Gap it shows |
|---|--------------|------|------|-------------|--------------|
```

Cite only sources you have actually read. A fake or unread reference counts as plagiarism.

4. Find your data source

Your project must use **real data**. Find a dataset that matches your problem on Kaggle, UCI Repository, Google Dataset Search or HDX, or collect your own with a survey.

- A few hundred records or more
- A licence that allows educational use
- No personal information that identifies real people

5. Create your GitHub repository

Name: **firstname_studentID_projectShortName**, for example `eric_12121_pharmaTrack`. Make it public (or add the instructor as a collaborator).

Your README must contain: project title, your name and ID, problem statement, objectives, data source, innovation, planned tables and PL/SQL features, and references.

Start with these folders (later phases add more): `docs/research/`, `docs/proposal/`, `docs/presentation/`, `data/`.

6. Write the proposal (maximum 2 pages)

Sections: Title · Problem Statement · Proposed Solution · Data Source · Planned Tables · Planned PL/SQL Features · Innovation · References (APA).

Format: Times New Roman 12 pt, headings 13–16 pt bold, line spacing 1.15, justified. Email it to eric.maniraguha@auca.ac.rw with your GitHub link.

7. Prepare the presentation (10–12 slides)

Slide	Content
1	Title, your name and ID, course, instructor, GitHub link
2	Background and problem: who is affected and why it matters
3	Motivation: why you chose this project
4	Objectives: one main objective, at least three specific ones
5	Research: existing solutions and their gaps
6	Data source: dataset, link, size, a few sample records
7	Proposed solution and innovation
8	Planned database: main tables and how they relate
9	PL/SQL approach: procedures, functions, cursors, triggers, packages, exceptions
10	Expected results: what you will demonstrate
11	Future improvements and limitations
12	References (APA)

> **Tip:** On slide 9, say **where** each feature is used, not just its name. Weak: “I will use triggers.” Strong: “A trigger blocks the sale of any expired drug.”

8. Innovation (8 marks)

Innovation is worth **8 of the 40 marks**. Make your database **interactive**: connect at least one tool that lets people see, use or protect the data beyond SQL Developer. Choose the option that best fits your problem.

Option	What you build	Example tools	Example
Analytics dashboard	A live dashboard on your Oracle data: KPIs, trends, filters	Power BI, Tableau, Excel Power Query	Monthly sales and expiring stock by district
User interface	A simple app where users enter and view data; the app calls your PL/SQL procedures	Oracle APEX, Python (Streamlit, Flask), Java, PHP, .NET	A clerk records a sale; the app calls `record_sale`
Security monitoring	Dashboards and alerts on your audit log, denied actions and database health	Grafana, Oracle Enterprise Manager, Kibana	Denied INSERT attempts per user per day
Smart automation	The database acts on its own on a schedule	DBMS_SCHEDULER, UTL_MAIL, REST APIs (ORDS)	A nightly job creates alerts and emails the manager
Advanced analytics	Scores, forecasts or predictions from your data	Oracle Machine Learning, Python with SQL, analytic SQL	A loan-default risk score for each borrower

Rules

- It must be **connected to your real Oracle database**. Static pictures or mock-ups do not count.
- Business rules stay in PL/SQL. The tool shows or uses the data; your procedures and triggers enforce the rules.
- You **demonstrate it live** in the final presentation.
- Document it in an `innovation/` folder: what it does, how to run it, and screenshots.

How the 8 marks are given

Criterion	Marks
Relevance: it helps solve your stated problem	2
Working connection to your Oracle data, shown live	3
Quality and ease of use	2
Documentation in GitHub	1

In Phase 0, name the option you plan and the tool you will use, on slide 7 and in your proposal. You will build it during the project and demonstrate it in Phase VII.

9. Phase 0 assessment and checklist

Area	Marks	Area	Marks
Problem and justification	15	PL/SQL approach	20
Research and data source	15	Expected results and future work	5
Objectives and innovation	15	Presentation and defence	10
Database design	20	Total	100

☐ Idea checked against the sheet and registered

☐ At least 6 sources in the research log, including a dataset

☐ GitHub repository named correctly with README

☐ Proposal emailed with GitHub link

☐ Presentation uploaded to GitHub

Proverbs 16:3

"Commit to the Lord whatever you do, and he will establish your plans."

This verse encourages us to **dedicate everything we do with full commitment and purpose**. It reminds us that proper planning, combined with hard work and focus, leads to better outcomes.

Excel with excellence!